

# Stabilizing MoE Routing via Stochastic Jitter

Reference: Fedus et al. (2022) — Switch Transformer

## 1. The Challenge: Routing Brittleness

Sparse MoE architectures suffer from training instability. If the router becomes overly confident early in training, it indefinitely starves certain experts of tokens, leading to model collapse.

## 2. Stochastic Perturbation (Training Only)

```
if is_training:
    # Inject uniform noise to force exploration across the expert pool
    noise = random_uniform(minval=1-eps, maxval=1+eps)
    router_logits = router_logits * noise

# Cast to FP32 for numerical stability before applying the Softmax activation
router_probs = softmax(to_float32(router_logits))
```

## 3. The Empirical Trade-Off

Jitter strictly enforces stability across all runs but acts as a regularization penalty on peak convergence.

| Routing Method       | Fraction of Stable Runs | Final Quality (Log Perplexity) |
|----------------------|-------------------------|--------------------------------|
| Baseline (No Jitter) | 4 / 6                   | -1.755 (Optimal)               |
| Input Jitter (1e-2)  | 3 / 3 (Stable)          | -1.777                         |
| Dropout (0.1)        | 3 / 3 (Stable)          | -1.822                         |